

Integrating Digital Twins and MPC for Sustainable Greenhouse Management in Smart Agriculture

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Abstract—This article introduces a digital twin framework for greenhouse climate management, utilizing model predictive control (MPC) to enhance smart farming practices and tackle the challenge of maintaining optimal conditions in diverse weather scenarios. The proposed MPC integrates dynamic system modeling, predictive control, and constraint management to regulate critical variables such as ambient temperature, solar radiation, and airflow. To validate the control framework, historical meteorological data from the city of Savona are used for four seasonal scenarios: winter, spring, summer, and autumn. The main contributions include designing and tuning the MPC, applying soft constraints via equal concern for relaxation for feasibility, and implementing scaling factors to ensure numerical stability. Simulation results demonstrate the controller's ability to maintain desired climatic conditions while meeting physical and operational constraints. This work highlights the potential of combining digital twin technology with advanced control systems to support sustainable and efficient smart farming.

Index Terms—Digital twin (DT), farm management systems, model predictive control (MPC), optimization, smart farming.

I. INTRODUCTION

THE rapid evolution of digital technologies has led to a significant transformation in agricultural production, enabling farmers to manage their operations more effectively and efficiently [1], [2]. Smart farming systems, based on integrating innovative technologies such as Internet of Things (IoT) [3], [4], cloud computing [5], [6], big data analytics [7], [8], machine learning [9], [10], and robotics [11], [12], have emerged as a natural evolution of precision agriculture [13]. As well as using precise location data, these systems integrate real-time context and event-based decision-making, supporting a comprehensive and dynamic approach to farm management [14].

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At the heart of this transformation is the concept of the digital twin (DT), a virtual counterpart to physical farm assets that reflect their behavior and status in real time [15], [16]. By leveraging sensor data [17], historical records [18], and predictive analytics [19], DTs enable farmers to remotely monitor, analyze, and control farm operations [20]. This paradigm shift disrupts physical processes from planning and control, eliminating the traditional constraints of time, location, and manual observation [21], [22]. Nowadays, farmers can assess and resolve potential problems through digital simulations and interventions, streamlining operations while improving sustainability and productivity [23].

Despite its potential, the application of DTs in intelligent agriculture is still in its infancy [24], [25]. The adoption of DTs in agriculture is expanding, even though the technology is still in its infancy in the sector. Several studies have explored its potential in various agricultural applications. For instance, Mitsanis et al. [26] highlighted the challenges of creating functional 3-D plant models for DTs, proposing a framework that integrates functional plant structure modeling to enhance yield and disease prediction. Similarly, Jeong et al. [27] investigated DTs in livestock farming, focusing on optimizing HVAC systems for energy efficiency while demonstrating the benefits through simulations using real-world pigsty datasets.

Furthermore, advances in controlled environment agriculture have been supported by DT architectures, such as those explored by [28], to optimize crop production and resource use through climate and treatment control strategies. In [29], it has developed a DT for smart irrigation systems, leveraging IoT platforms, and real-time simulations to assess and improve water use and operational efficiency. In the same context, Manocha et al. [30] proposed a smart irrigation framework using DTs and IoT for efficient water management, highlighting its potential for reducing water consumption and optimizing irrigation practices. In addition, a framework integrating IoT to improve vertical farming by improving productivity, environmental adaptation, and energy savings while enabling continuous monitoring and optimization of agrifood lifecycles was presented by Monteiro et al. [31]. Indeed, a new architecture exploiting cloud, fog, edge computing, and multiagent systems has been introduced by Kalyani et al. [32] to enhance smart agriculture, addressing challenges such as resource allocation and scheduling and demonstrating the integration of DTs into advanced agricultural applications.

In urban agriculture, Clementi et al. [33] proposed a GIS-based DT model for real-time solar radiation mapping, optimizing crop productivity and resilience in urban environments, with validation in Milan and potential application in Singapore. In the same context, Ghandar et al. [34] extended the scope by applying DTs to urban agriculture through aquaponics, showing how such systems can reduce environmental burdens and improve production logistics using adaptive decision support tools. In vertical indoor agriculture, Mengi et al. [35] developed a DT framework for optimizing LED-based illumination, employing genetic algorithms to enhance energy efficiency and productivity in hydroponic systems. Slob et al. [36] investigated immersive technologies in digital greenhouse twins, focusing on simulation sickness and user interaction, finding correlations between prior gaming experience, playability, and user ratings. In [37], a DT for rice cultivation was explored, using an ontology-based knowledge base to automate resource scheduling and decision-making, proving a better return on investment of precision agriculture in pilot tests.

In terms of agricultural machinery, Nguyen et al. [38] demonstrated the use of model predictive control (MPC) in four-wheel-steer vehicles to optimize maneuverability under the complex constraints of agricultural fields. All these studies highlight the transformative potential of DTs to improve productivity, resource efficiency, and sustainability in a variety of agricultural fields.

The examined studies highlight significant advances in the application of DTs to smart agriculture, including resource optimization, climate control, and improved decision-making. However, a limited focus, such as addressing a single aspect like irrigation or energy efficiency, constricts many of these studies, and they often lack integration between real-time control systems and comprehensive performance assessments. Moreover, few works address the practical challenges of implementing large-scale digital twins, such as system reliability, computational efficiency, and adaptability to varying environmental conditions.

This article addresses these limitations by presenting a fully integrated DT framework for greenhouse management, exploiting MPC to achieve precise and dynamic control of key environmental variables, such as ventilation, shading, and heating, thus guaranteeing energy efficiency and crop health. In contrast to existing works, this study evaluates the system in various seasonal scenarios, offering a broader understanding of its operational robustness. In addition, the integration of environmental and economic impact analyses ensures the model's practicality and sustainability, making it a valuable contribution to the field of smart farming.

The rest of this article is organized as follows. Section II describes the system architecture, including the mathematical model. Sections III and IV detail the MPC-based methodology employed in the proposed framework, as well as present simulation results and performance evaluation under varying seasonal conditions. Sections V, VI, and VII present the physical domain, the cybernetic domain, and the graphical user interface (GUI), respectively. A discussion of the results, including the economic and operational impacts of the proposed model, is given in Section VIII. Finally, Section IX concludes this article.

II. SYSTEM ARCHITECTURE AND MODELING

The methodology adopted in this research follows a design-oriented approach aimed at developing a DT framework for a greenhouse control system. The overall approach is organized into the following three distinct phases:

- 1) model design;
- 2) MPC controller design;
- 3) DT framework development.

The first phase, *model design*, commenced with a comprehensive analysis of the greenhouse prototype located at the Savona Campus, University of Genova [39]. A mathematical model of the greenhouse was developed to capture the system's behavior, focusing on the key inputs and outputs. This model served as the foundation for subsequent phases, defining the operational boundaries and establishing the dynamic behavior that would inform both the control design and the DT development. In the second phase, *MPC controller design*, the greenhouse model was employed to design an MPC controller. The design process involved specifying the controller parameters and tuning its performance through simulation. The input/output ranges of the MPC were fully defined, facilitating its interaction with the DT framework in the later stages. The controller was designed to optimize the internal climate of the greenhouse by adjusting inputs, such as the heating/cooling system and window apertures, based on predicted environmental conditions. Finally, the third phase involved the development of the *DT framework*. The framework is structured into three interconnected domains: the *cyber domain*, the *physical domain*, and the *GUI*. Initially, the cyber domain was designed to meet the input and output requirements of the MPC controller, ensuring seamless integration with MATLAB for real-time data processing and simulation. The physical domain was then modeled to interface with the cyber domain, incorporating necessary sensors and actuators to capture real-world data. Subsequently, a user-friendly GUI was developed, allowing operators to monitor and control the greenhouse system through the DT, thereby reflecting real-time data and operational states. The methodology outlined above provides a structured approach to developing a DT framework that integrates the greenhouse system's cyber and physical components. This integration enables real-time simulation, monitoring, and control of the greenhouse environment, facilitating more efficient agricultural operations management.

A. Mathematical Model

The mathematical model of the greenhouse system was derived using a state-space representation [39], [40], simplifying the system dynamics for control and simulation purposes. Fig. 1 represents the states where control inputs and external inputs are gathered. The system states are represented by the following equation:

$$\dot{x} = f(t, x, u, d) \quad (1)$$

where t time, x represents the system states, u are the control inputs, and d are the external inputs. The nonlinear function f captures the dynamics of the greenhouse, taking into account heat transfer processes, ventilation, and external environmental conditions.

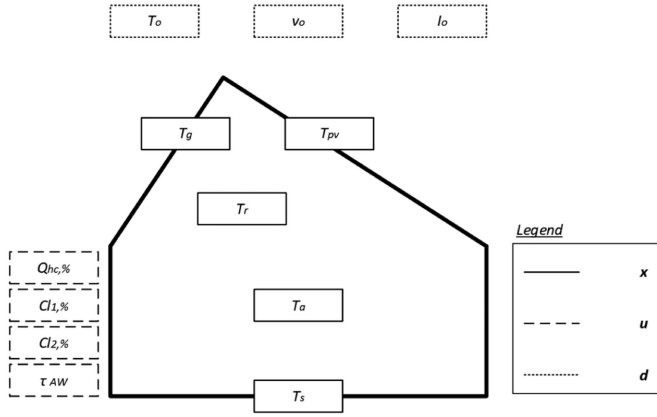


Fig. 1. Greenhouse model: States x (solid box), control inputs u (dashed box), and external inputs d (dotted box) in the model.

TABLE I
PARAMETERS USED IN THE GREENHOUSE MODEL

Symbol	Description	Unit
T_g	Temperature of glass roof	K
T_{pv}	Temperature of PV module roof	K
T_r	Indoor air temperature below roof	K
T_a	Indoor air temperature	K
T_s	Soil temperature	K
Q_{hc}	Heating/Cooling power	W
$CI1\%$	Window aperture (lee-side)	[0,1]
$CI2\%$	Window aperture (windward-side)	[0,1]
τ_{AW}	Awning extension	[0,1]
T_o	Outdoor air temperature	K
I_o	Outdoor shortwave solar radiation	W/m ²
v_o	Outdoor wind speed	m/s
ρ_a	Air density (ambient)	kg/m ³
c_{pa}	Specific heat capacity of air	J/kg·K
V_a	Volume of air (ambient)	m ³
ρ_r	Air density (roof)	kg/m ³
c_{pr}	Specific heat capacity of air (roof)	J/kg·K
V_r	Volume of air (roof)	m ³
ρ_g	Density of glass	kg/m ³
c_{pg}	Specific heat capacity of glass	J/kg·K
A_g	Area of glass roof	m ²
V_g	Volume of glass roof	m ³
ρ_{pv}	Density of PV module	kg/m ³
$c_{p,pv}$	Specific heat capacity of PV module	J/kg·K
A_{pv}	Area of PV module roof	m ²
V_{pv}	Volume of PV module roof	m ³
ρ_s	Soil density	kg/m ³
c_{ps}	Specific heat capacity of soil	J/kg·K
A_s	Soil area	m ²
V_s	Soil volume	m ³
α_{AB}	Heat transfer coefficient between A and B (U-factor)	W/m ² ·K
$\eta_{A_{Ts}}$	Shortwave radiation absorption coefficient for A	-
λ	Thermal conductivity	W/m·K
d	Thickness of material	m
F_{win}	Airflow through windows	m ³ /s
$F_{a,r}$	Airflow from lower to upper part of the greenhouse	m ³ /s
g	Gravitational constant	m/s ²

The model was simplified to focus on key thermal processes, such as the heat exchange between indoor air, the roof, and the soil. All the parameters and variables used in the greenhouse model, along with their descriptions and units, are summarized in Table I.

B. Temperature of Indoor Air Temperature (K/s)

$$\frac{dT_a}{dt} = \frac{-Q_{ar} - Q_{as} + Q_{hc}}{\rho_a c_{p,a} V_a} \quad (2)$$

where T_a is the indoor air temperature (K/s), Q_{ar} is the heat exchange between indoor air and the roof, Q_{as} is the heat exchange with the soil, and Q_{hc} is the heating/cooling input. ρ_a is the average density of air (kg/m³), $c_{p,a}$ is the specific heat capacity of air (J/kg · K), and V_a is the volume of air (ambient) (m³).

C. Temperature of Soil (K/s)

$$\frac{dT_s}{dt} = \frac{Q_{as}}{\rho_s c_{p,s} V_s} \quad (3)$$

where T_s is the temperature of the soil (K), and ρ_s represents the density of the soil (kg/m³), $c_{p,s}$ is the specific heat capacity of the soil (J/kg · K), and V_s is the volume of the soil (m³).

D. Temperature of Indoor Air Below the Roof (K/s)

$$\frac{dT_r}{dt} = \frac{Q_{ar} - Q_{ro} - Q_{rg} - Q_{r,pv}}{\rho_r c_{p,r} V_r} \quad (4)$$

where T_r is the temperature of the indoor air below the roof (K), Q_{ro} is the heat flow between the roof and the outdoor (W), Q_{rg} is the heat flow between the roof and the glass (W), and $Q_{r,pv}$ is the heat flow between the roof and the PV module (W), ρ_r is the density of the air below the roof (kg/m³), $c_{p,r}$ is the specific heat capacity of the air below the roof (J/kg · K), and V_r is the volume of the air below the roof (m³).

E. Temperature of Indoor Side of the Roof (Glass) (K/s)

$$\frac{dT_g}{dt} = \frac{Q_{radg} + Q_{rg} - Q_{go}}{\rho_g c_{p,g} V_g} \quad (5)$$

where T_g is the temperature of the indoor side of the glass roof (K), Q_{radg} is the radiative heat flow to the glass (W), Q_{go} is the heat flow between the glass and the outdoor (W), ρ_g represents the density of the glass (kg/m³), $c_{p,g}$ is the specific heat capacity of the glass (J/kg · K), and V_g is the volume of the glass (m³).

F. Temperature of Indoor Side of the Roof (PV Module) (K/s)

$$\frac{dT_{pv}}{dt} = \frac{Q_{radpv} + Q_{r,pv} - Q_{pv0}}{\rho_{pv} c_{p,pv} V_{pv}} \quad (6)$$

where T_{pv} is the temperature of the indoor side of the PV module (K), Q_{radpv} is the radiative heat flow to the PV module (W), Q_{pv0} is the heat flow between the PV module and the outdoor (W), ρ_{pv} is the density of the PV module (kg/m³), $c_{p,pv}$ is the specific heat capacity of the PV module (J/kg · K), and V_{pv} is the volume of the PV module (m³).

TABLE II
PARAMETER VALUES USED IN THIS MODEL

Material	ρ (kg/m ³)	c_p (J/kg·K)	A (m ²)	V (m ³)	$c_p \cdot \rho \cdot V$ (10 ⁶ J/K)
Air (a)	1.29	1000	25	50	0.06
Air (r)	1.29	1000	-	27.94	0.04
Roof (g)	2500	840	12.7	0.20	0.43
PV mod. (PV)	1069	800	22	0.09	0.11
Soil (s)	1600	800	25	12.5	16

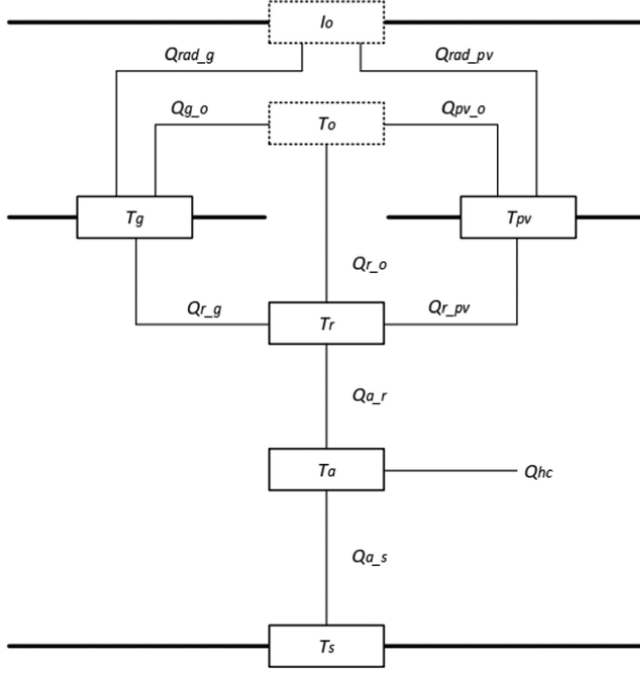


Fig. 2. Thermal model: States x (solid box) and external inputs d (dotted box) in the thermal model.

Table II shows the parameters of the main materials used in this model [39], [40].

1) *Thermal Model*: The thermal model has been improved by incorporating the absorption of shortwave radiation, the effects of natural ventilation, and a simplified representation of the heating and cooling systems. The complete set of state equations, control inputs, and external variables as detailed in [39], provides a solid basis for simulating the greenhouse's dynamics and optimizing its internal climate control system. Fig. 2 provides a visual representation of the thermal model, where solid boxes represent states, dashed boxes represent external inputs, and lines represent heat transfer between states by convection, absorption of shortwave radiation, or conduction. Due to the limitations of real-time data processing, this model presents a simplified version of the real system.

2) *Convection*: Convection is the heat transfer between a solid and a fluid. The heat transfer Q_{A_B} from A to B is described by Newton's law of cooling, as presented in the following equation:

$$Q_{A_B} = A_{A_B} \alpha_{A_B} (T_A - T_B) \quad (7)$$

where A_{A_B} (m²) is the surface area during the heat transfer, α_{A_B} (W/m² K) is the heat transfer coefficient between A and

TABLE III
SHORTWAVE RADIATION VALUES

Q_{rad_A}	A_A	$\eta_{A_{Is}}$
Q_{rad_g}	A_g	$\eta_{g_{Is}} = 0.56(1 - \tau_{AW})$
$Q_{\text{rad}_{pv}}$	A_{pv}	$\eta_{pv_{Is}} = 0.85$

TABLE IV
CONDUCTION VALUES

Q_{A_B}	A_{A_B}	λ	d	T_A	T_B
Q_{g_o}	A_g	0.9	0.016	T_g	T_o
Q_{pv_o}	A_{pv}	148	0.004	T_{pv}	T_o

B (U-factor), and T_A and T_B (K) are the temperatures of A and B, respectively. Heat transfer due to convection is given to all surfaces.

The heat exchange between the ambient indoor temperature and the air below the roof is defined by the following equation:

$$Q_{a_r} = \rho_a c_{p,a} F_{a_r} (T_a - T_r) \quad (8)$$

where F_{a_r} (m³/s) is the flow of air from below to above, and T_a and T_r the indoor temperatures of the lower and upper parts of the greenhouse, respectively.

Heat exchange between indoor and outdoor temperatures through ventilation is defined by the following equation:

$$Q_{r_o} = \rho_a c_{p,a} F_{\text{win}} (T_r - T_o) \quad (9)$$

where F_{win} (m³/s) is the flow of air through the greenhouse, and T_o is the outdoor temperature. The airflow variables, F_{win} and F_{a_r} , are described in the ventilation model.

3) *Shortwave Radiation Absorption*: Shortwave radiation encompasses all wavelengths in the shortwave range, whether received directly or indirectly from the sun. In this case, the heat Q_{rad_A} absorbed by object A is defined by the following equation:

$$Q_{\text{rad}_A} = A_A \eta_{A_{Is}} I_o \quad (10)$$

where A_A (m²) is the heat transfer surface, $\eta_{A_{Is}}$ (–) is the shortwave radiation absorption coefficient for A, and I_o (W/m²) is the external shortwave radiation. Table III shows the heat transfer due to shortwave radiation for all surfaces [39]: Where τ_{AW} (–) represents the control input for extending the reflecting curtains.

4) *Conduction*: Conduction refers to the heat transfer Q_{A_B} between locations A and B within a homogeneous medium, as described by the following equation:

$$Q_{A_B} = A_{A_B} \frac{\lambda}{d} (T_A - T_B) \quad (11)$$

where λ (W/m · K) is the thermal conductivity of the homogeneous medium, and d is the material thickness (m). Table IV outlines the conduction values for heat transfer across various surfaces in the greenhouse system. It details the heat transfer quantities Q_{A_B} , surface areas A_{A_B} , thermal conductivities λ , material thicknesses d , and temperatures T_A and T_B for each pair of surfaces. Specifically, it provides values for heat conduction from the greenhouse glass (Q_{g_o}) and photovoltaic panels (Q_{pv_o}) to the outside environment, highlighting the

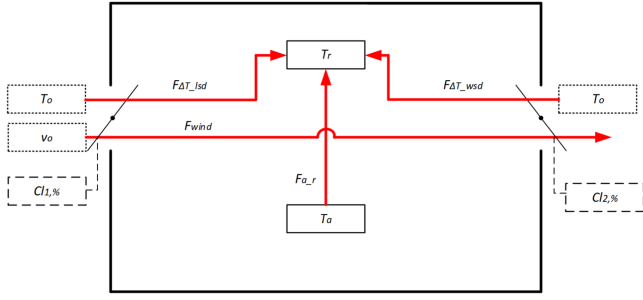


Fig. 3. Ventilation model; states (solid boxes), external inputs (dotted boxes), control inputs (dashed boxes), and airflows (red arrows).

TABLE V
CHARACTERISTICS OF THE WINDOWS

Type of window	h_{win} (m)	w_{win} (m)	n_{win} (-)
Window A	0.5	0.5	2

distinct thermal properties and thicknesses of each material. The thickness values (d) are sourced from [39], while the thermal conductivity values (λ) are based on data from [41].

5) *Ventilation Model*: The ventilation model is shown in Fig. 3, and it considers two main airflows: an indoor airflow driven by temperature differences and an external airflow entering the greenhouse through the windows.

The window characteristics, such as height, width, and number, are provided in Table V.

Ventilation in the greenhouse is initiated by the opening of windows. Following the equations presented in [42], the simplified model accounts for the wind-induced component and the temperature difference between the indoor and outdoor environments, while neglecting the effects of air leakage. The airflow through the windows, F_{win} , is given as

$$F_{win} = \sqrt{F_{wind}^2 + (F_{\Delta T_{lzd}} + F_{\Delta T_{wsd}})^2} \quad (12)$$

where F_{wind} (m^3/s) is the wind-induced component, and $F_{\Delta T_{lzd}}$ and $F_{\Delta T_{wsd}}$ (m^3/s) represent the components based on the temperature difference between the indoor and outdoor sides on the lee and windward sides, respectively.

The wind-induced airflow component, F_{wind} , is described by

$$F_{wind} = n_{win}(f_{lzd} + f_{wsd})v_o w_{win} h_{win} \quad (13)$$

where f_{lzd} and f_{wsd} (-) are ventilation functions for the lee and windward sides, w_{win} and h_{win} (m) are the width and height of the windows, n_{win} (-) is the number of windows, and v_o (m/s) is the wind speed.

The ventilation functions, f_{lzd} and f_{wsd} , are given by

$$f_{lzd} = c_{lzd3}C_{l1,\%}^3 - c_{lzd2}C_{l1,\%}^2 + c_{lzd1}C_{l1,\%} + c_{lzd0} \quad (14)$$

$$f_{wsd} = c_{wsd3}C_{l2,\%}^3 - c_{wsd2}C_{l2,\%}^2 + c_{wsd1}C_{l2,\%} + c_{wsd0} \quad (15)$$

where $C_{l1,\%} \in [0, 1]$ and $C_{l2,\%} \in [0, 1]$ represent the aperture percentage of the windows on both sides. The coefficients c_{wsdx} and c_{lzd3} are listed in Table VI.

The components $F_{\Delta T_{lzd}}$ and $F_{\Delta T_{wsd}}$, representing the temperature-driven airflow on the lee and windward sides, are

TABLE VI
VENTILATION COEFFICIENTS FOR LEE- AND WINDWARD SIDES

Side	c_3	c_2	c_1	c_0
Lee	$1.22 \cdot 10^{-2}$	$3.77 \cdot 10^{-2}$	$4.71 \cdot 10^{-2}$	$1.26 \cdot 10^{-4}$
Windward	$2.42 \cdot 10^{-2}$	$4.48 \cdot 10^{-2}$	$7.08 \cdot 10^{-2}$	$5.91 \cdot 10^{-4}$

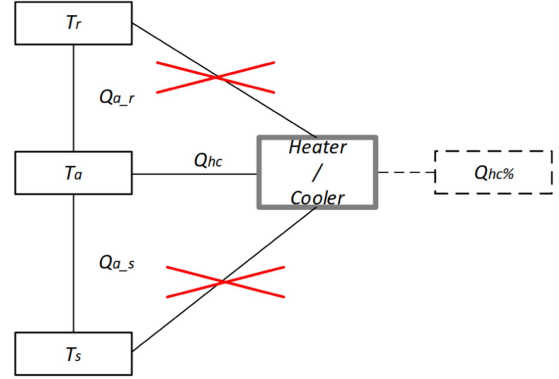


Fig. 4. Heating and cooling model: States (solid box) and control inputs (dashed box).

defined by

$$F_{\Delta T_{lzd}} = n_{win} \frac{c_w}{3} w_{win} \sqrt{g \frac{|T_o - T_r|}{T_o}} L_{win,lzd}^{1.5} \quad (16)$$

$$F_{\Delta T_{wsd}} = n_{win} \frac{c_w}{3} w_{win} \sqrt{g \frac{|T_o - T_r|}{T_o}} L_{win,wsd}^{1.5} \quad (17)$$

where w_{win} (m) is the window width, $c_w = 0.6$ (-) is the discharge coefficient, $g = 9.81 \text{ m/s}^2$ is the gravitational constant, and $L_{win,lzd}$ and $L_{win,wsd}$ (m) are the vertical projections of the window openings.

Finally, the indoor airflow, $F_{a,r}$, which circulates from the lower section of the greenhouse to the upper part beneath the roof due to temperature differences, is described by

$$F_{a,r} = v_{a,r} A_a \quad (18)$$

where A_a (m^2) is the surface area for heat exchange and $v_{a,r}$ (m/s) is the air exchange rate evaluated 0.05 in according to [40].

6) *Reflecting Curtains Model*: To reduce the influence of shortwave radiation on the indoor environment, reflective curtains are installed on the glass-side of the roof. The roof's shortwave radiation absorption coefficient is expressed as

$$\eta_{g,JS} = 0.56(1 - \tau_{AW}) \quad (19)$$

where $\tau_{AW} \in [0, 1]$ represents the control input governing the extension of the curtains. The greenhouse effect resulting from the longwave radiation absorption due to the reflective curtains is not considered.

7) *Heating/Cooling System Model*: The heating and cooling system designed for the greenhouse is a simplified representation of the actual system. It interacts solely with the indoor temperature state, T_a , while being isolated from other heat fluxes. Fig. 4 illustrates the structure of this heating/cooling system.

III. PREDICTIVE CONTROL OF THE MIMO SYSTEM

A. Formulation of the Model

The MPC model formulation follows the methodology presented in [43]. The current system has $m = 7$ inputs, $q = 5$ outputs (only one of which is measured), and $n_1 = 5$ states. For effective control, the number of outputs should be less than or equal to the number of inputs to manage each measured output (MO) independently and avoid steady-state errors.

Neglecting plant noise and disturbances, the general formulation of the MIMO DMPC is described by

$$x_m(k+1) = A_m x_m(k) + B_m u(k) \quad (20)$$

$$y(k) = C_m x_m(k) \quad (21)$$

defining $\Delta x_m(k) = x_m(k) - x_m(k-1)$ and $\Delta u(k) = u(k) - u(k-1)$, we obtain

$$\Delta x_m(k+1) = A_m \Delta x_m(k) + B_m \Delta u(k) \quad (22)$$

$$\Delta y(k+1) = C_m \Delta x_m(k+1) \quad (23)$$

choosing a new state-space vector $x(k) = [\Delta x_m(k)^T y(k)^T]^T$, we can express the augmented state-space vector as

$$\begin{bmatrix} \Delta x_m(k+1) \\ y(k+1) \end{bmatrix} = \begin{bmatrix} A_m & 0_m^T \\ C_m A_m & I_q \end{bmatrix} \begin{bmatrix} \Delta x_m(k) \\ y(k) \end{bmatrix} + \begin{bmatrix} B_m \\ C_m B_m \end{bmatrix} \Delta u(k) \quad (24)$$

$$y(k) = \begin{bmatrix} 0_m & I_q \end{bmatrix} \begin{bmatrix} \Delta x_m(k) \\ y(k) \end{bmatrix} \quad (25)$$

where I_q is the identity matrix of size $q \times q$, and 0_m is a $q \times n_1$ zero matrix. Matrices A_m , B_m , and C_m have dimensions $n_1 \times n_1$, $n_1 \times m$, and $q \times n_1$, respectively. For simplicity, we denote the state-space model as

$$x(k+1) = Ax(k) + B\Delta u(k) \quad (26)$$

$$y(k) = Cx(k). \quad (27)$$

B. Prediction of State and Output Variables

Following model formulation, the next step is to predict the plant output by adjusting the future control signal. Let k_i be the current time, N_p the prediction horizon, and N_c the control horizon.

The control trajectory for future steps is represented as

$$\Delta u(k_i), \Delta u(k_i+1), \dots, \Delta u(k_i+N_c-1). \quad (28)$$

The future state variables are represented as

$$x(k_i+1|k_i), x(k_i+2|k_i), \dots, x(k_i+N_p|k_i). \quad (29)$$

These state variables are computed iteratively using the control parameters

$$x(k_i+1|k_i) = Ax(k_i) + B\Delta u(k_i)$$

$$x(k_i+2|k_i) = Ax(k_i+1|k_i) + B\Delta u(k_i+1)$$

$$= A^2x(k_i) + AB\Delta u(k_i) + B\Delta u(k_i+1)$$

$$\vdots$$

$$\begin{aligned} x(k_i+N_p|k_i) &= A^{N_p}x(k_i) + A^{N_p-1}B\Delta u(k_i) \\ &\quad + A^{N_p-2}B\Delta u(k_i+1) + \dots + \\ &\quad A^{N_p-N_c}B\Delta u(k_i+N_c-1). \end{aligned} \quad (30)$$

With these state variables, the predicted output values can be obtained by substitution

$$\begin{aligned} y(k_i+1|k_i) &= CAx(k_i) + CB\Delta u(k_i) \\ y(k_i+2|k_i) &= CA^2x(k_i) + CAB\Delta u(k_i) + CB\Delta u(k_i+1) \\ y(k_i+3|k_i) &= CA^3x(k_i) + CA^2B\Delta u(k_i) \\ &\quad + CAB\Delta u(k_i+1) + CB\Delta u(k_i+2) \\ &\vdots \end{aligned} \quad (31)$$

$$\begin{aligned} y(k_i+N_p|k_i) &= CA^{N_p}x(k_i) + CA^{N_p-1}B\Delta u(k_i) \\ &\quad + CA^{N_p-2}B\Delta u(k_i+1) + \dots \\ &\quad + CA^{N_p-N_c}B\Delta u(k_i+N_c-1). \end{aligned} \quad (32)$$

The subsequent step involves defining the vectors $\mathbf{Y}$ and $\Delta \mathbf{U}$ as illustrated in

$$\Delta \mathbf{U} = [\Delta u(k_i)^T \Delta u(k_i+1)^T \dots \Delta u(k_i+N_c-1)^T]^T \quad (33)$$

$$\mathbf{Y} = [y(k_i+1|k_i)^T y(k_i+2|k_i)^T \dots y(k_i+N_p|k_i)^T]^T. \quad (34)$$

The vectors $\Delta \mathbf{U}$ and $\mathbf{Y}$ shall be gathered in a matrix compact form described in

$$\mathbf{Y} = Fx(k_i) + \Phi \Delta \mathbf{U} \quad (35)$$

where

$$F = \begin{bmatrix} CA \\ CA^2 \\ CA^3 \\ \vdots \\ CA^{N_p} \end{bmatrix} \quad (36)$$

$\Phi =$

$$\begin{bmatrix} CB & 0 & 0 & \dots & 0 \\ CAB & CB & 0 & \dots & 0 \\ CA^2B & CAB & CB & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ CA^{N_p-1}B & CA^{N_p-2}B & CA^{N_p-3}B & \dots & CA^{N_p-N_c}B \end{bmatrix}. \quad (37)$$

C. Optimization

The objective of the predictive control system is to align the predicted output as closely as possible with the set-point (SP) signal $r(k_i)$ at each sampling time k_i . To achieve this, the controller aims to determine the vector ΔU that minimizes the error between the SP signal and the predicted output.

The cost function J , representing the control objective, is expressed as

$$J = (R_s - Y)^T (R_s - Y) + \Delta U^T \bar{R} \Delta U \quad (38)$$

where

R_s data vector containing all the SP information;

$\bar{R}$ diagonal matrix used to tune the closed-loop performance, given by

$$\bar{R} = r_w I_{N_c} \quad (39)$$

where $r_w \geq 0$, and I_{N_c} is the identity matrix of size $N_c \times N_c$.

The incremental optimal control within a single optimization window is determined by the optimal solution for the control signal

$$\Delta U = (\Phi^T \Phi + \bar{R})^{-1} (\Phi^T \bar{R}_s r(k_i) - \Phi^T F x(k_i)). \quad (40)$$

D. Formulation of Constraints

The predictive control problem is formulated as an optimization problem that incorporates design constraints. These constraints are categorized as follows.

- 1) Constraints on the incremental variation of the control variable: $\Delta u^{\max}$, $\Delta u^{\min}$.
- 2) Constraints on the amplitude of the control variable: $u^{\max}$, $u^{\min}$.

Assuming unconstrained output variables, these constraints are specified independently for each input.

The upper and lower bounds for the incremental variation of the control variable are defined as

$$\begin{bmatrix} \Delta u_1^{\max} & \Delta u_2^{\max} & \dots & \Delta u_m^{\max} \end{bmatrix}.$$

The lower limits for the incremental variation are denoted as

$$\begin{bmatrix} \Delta u_1^{\min} & \Delta u_2^{\min} & \dots & \Delta u_m^{\min} \end{bmatrix}.$$

The variables representing the rate of change are defined as

$$\begin{bmatrix} \Delta u_1^{\min} \leq \Delta u_1(k) \leq \Delta u_1^{\max} \\ \Delta u_2^{\min} \leq \Delta u_2(k) \leq \Delta u_2^{\max} \\ \vdots \\ \Delta u_m^{\min} \leq \Delta u_m(k) \leq \Delta u_m^{\max} \end{bmatrix}. \quad (41)$$

The upper limits for the control signal are defined as

$$\begin{bmatrix} u_1^{\max} & u_2^{\max} & \dots & u_m^{\max} \end{bmatrix}.$$

The lower limits for the control signal are denoted as

$$\begin{bmatrix} u_1^{\min} & u_2^{\min} & \dots & u_m^{\min} \end{bmatrix}.$$

Thus, the amplitude of the control signal is constrained as follows:

$$\begin{bmatrix} u_1^{\min} \leq u_1(k) \leq u_1^{\max} \\ u_2^{\min} \leq u_2(k) \leq u_2^{\max} \\ \vdots \\ u_m^{\min} \leq u_m(k) \leq u_m^{\max} \end{bmatrix}. \quad (42)$$

To incorporate these constraints into the predictive control framework, they are reformulated as linear inequalities by parameterizing the constrained variables in terms of ΔU . These constraints are applied to all future sampling instances, ensuring that every constraint is expressed as a function of ΔU

$$\begin{aligned} & \begin{bmatrix} u(k_i) \\ u(k_i + 1) \\ u(k_i + 2) \\ \vdots \\ u(k_i + N_c - 1) \end{bmatrix} \\ &= \begin{bmatrix} I \\ I \\ I \\ \vdots \\ I \end{bmatrix} u(k_i - 1) + \begin{bmatrix} I & 0 & 0 & \dots & 0 \\ I & I & 0 & \dots & 0 \\ I & I & I & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ I & I & I & \dots & I \end{bmatrix} \Delta U \\ & \begin{bmatrix} \Delta u(k_i) \\ \Delta u(k_i + 1) \\ \Delta u(k_i + 2) \\ \vdots \\ \Delta u(k_i + N_c - 1) \end{bmatrix}. \end{aligned} \quad (43)$$

Rewriting the above expression into a compact matrix form, where C_1 and C_2 correspond to the appropriate matrices, the constraints for the control movement are described in the following equations, respectively:

$$-(C_1 u(k_i - 1) + C_2 \Delta U) \leq -U^{\min} \quad (44)$$

$$(C_1 u(k_i - 1) + C_2 \Delta U) \leq U^{\max} \quad (45)$$

where $U^{\min}$ and $U^{\max}$ are column vectors with N_c elements of $u^{\min}$ and $u^{\max}$. For the incremental variation of the signals, we have

$$-\Delta U \leq -\Delta U^{\min} \quad (46)$$

$$\Delta U \leq \Delta U^{\max} \quad (47)$$

where $\Delta U^{\min}$ and $\Delta U^{\max}$ are column vectors with N_c elements of $\Delta u^{\min}$ and $\Delta u^{\max}$, respectively.

The MPC with hard constraints is formulated as determining ΔU that reduces the expression given by

$$\begin{aligned} J &= (R_s - F x(k_i))^T (R_s - F x(k_i)) \\ &\quad - 2\Delta U^T \Phi^T (R_s - F x(k_i)) + \Delta U^T (\Phi^T \Phi + R) \Delta U. \end{aligned} \quad (48)$$

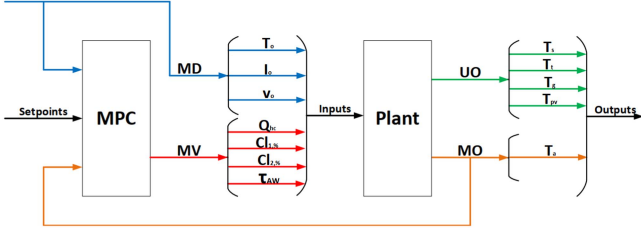


Fig. 5. MPC structure block diagram.

Subject to the inequality constraints

$$\begin{bmatrix} M_1 \\ M_2 \end{bmatrix} \Delta U \leq \begin{bmatrix} N_1 \\ N_2 \end{bmatrix} \quad (49)$$

where the data matrices are

$$M_1 = \begin{bmatrix} -C_2 \\ C_2 \end{bmatrix}, \quad M_2 = \begin{bmatrix} -I \\ I \end{bmatrix} \quad (50)$$

$$N_1 = \begin{bmatrix} -U^{\min} + C_1 u(k_i - 1) \\ U^{\max} - C_1 u(k_i - 1) \end{bmatrix}, \quad N_2 = \begin{bmatrix} -\Delta U^{\min} \\ \Delta U^{\max} \end{bmatrix}. \quad (51)$$

IV. MPC SPECIFICATION

A. MPC Structure

The structure of the MPC defines the types and sizes of signals utilized by the MPC controller. The block diagram of the MPC structure is presented in Fig. 5.

1) *Plant Inputs*: Plant inputs are the independent variables that influence the plant. These inputs can be categorized into the following three types.

- 1) *Measured disturbances (MD)*: Variables that the controller cannot adjust but are utilized for feedforward compensation. Examples include T_o , I_o , and v_o , which are estimated using weather forecasts and real-time measurements.
- 2) *Manipulated variables (MV)*: Variables that the controller can adjust to achieve its objectives. The MV in the current MPC design are $Q_{hc\%}$, $Cl_{1\%}$, $Cl_{2\%}$, and τ_{AW} .
- 3) *Unmeasured disturbances (UD)*: Variables that are not directly known to the controller. In the present MPC design, there are no UD.

2) *Plant Outputs*: Plant outputs are the dependent variables that are either controlled or monitored. These outputs are classified into the following two types.

- 1) *MOs*: Variables used by the controller for estimating unmeasured variables and providing feedback on the effectiveness of control actions. The only MO in the current design is T_a .
- 2) *Unmeasured outputs (UO)*: variables estimated by the controller but not directly measured via feedback. The UO in this design includes the signals T_s , T_r , T_g , and T_{pv} .

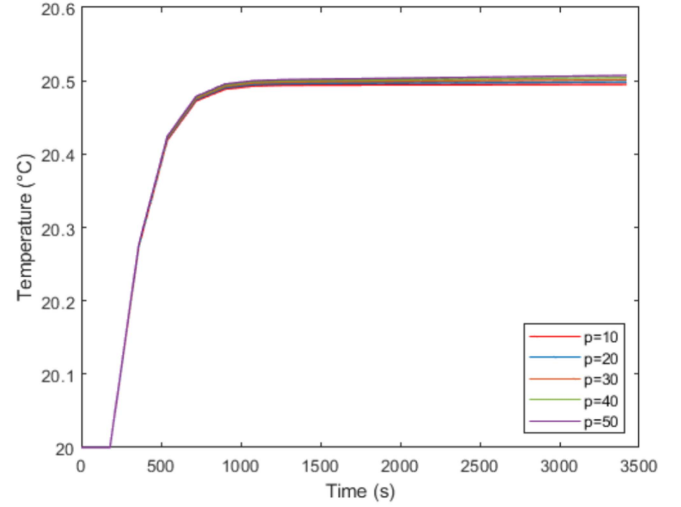


Fig. 6. Comparison of prediction horizons: Ambient temperature response to a unit step SP (Step time = 180 s, control horizon = 3).

TABLE VII
SAMPLE TIME AND HORIZONS VALUES

Parameter	Value [s]
Sample time	180
Prediction horizon	50
Control horizon	5

B. MPC Design

During the MPC design, the first parameter to set is the sample time. This parameter should not be adjusted after tuning. Reducing the sample time improves disturbance rejection but increases computational effort. Thus, the optimal sample time balances performance and computational efficiency.

For process control, sample times are typically much greater than 1 second. Fig. 6 compares the ambient temperature response (T_a) to a unitary step in the SP for different sample times. The system response slows down as the sampling time increases. According to the controller design guidelines provided by MATLAB [44], the sampling time should be set between 10% and 25% of the minimum desired closed-loop response time. Since the desired closed-loop response time is 1800 s, a sampling time of 180 s is chosen. For the prediction horizon, best practice recommends selecting its value early in the controller design process and keeping it constant while setting other parameters. The prediction horizon should be chosen to ensure that the controller remains internally stable and effectively anticipates constraint violations. The control horizon is kept significantly smaller than the prediction horizon to promote faster computations and internal stability. The chosen values are summarized in Table VII.

In order to prevent the plant's physical limits from being exceeded, design constraints are put in place. Since there is no need to place limits on the output variables, constraints are only applied to the MVs. Fig. 7 compares unconstrained (green) and constrained (blue) control, where the unconstrained control violates upper limits (red).

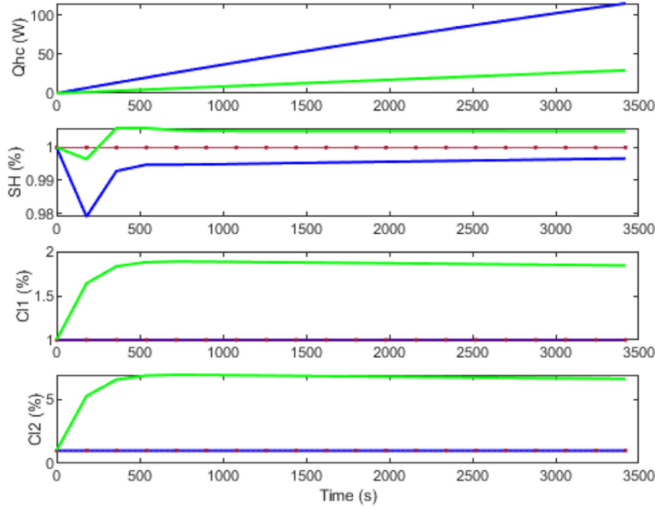


Fig. 7. Unconstrained (green) versus constrained control (blue): MVs review.

TABLE VIII
SPECIFICATION OF CONSTRAINTS

MV	$u^{\max}$	$u^{\min}$	$\Delta u^{\max}$	$\Delta u^{\min}$
Q_{hc}	5000	-5000	200	-200
τ_{AW}	1	0	0.1	-0.1
CI1%	1	0	0.5	-0.5
CI2%	1	0	0.5	-0.5

TABLE IX
SPECIFICATION OF ECR PARAMETERS

MV	$u^{\max}$	$u^{\min}$	$\Delta u^{\max}$	$\Delta u^{\min}$
Q_{hc}	0	0	1	1
τ_{AW}	0	0	1	1
CI1%	0	0	0.1	0.1
CI2%	0	0	0.1	0.1

The constraints are listed in Table VIII. Incremental limits are treated as soft constraints by setting equal concern for relaxation (ECR) parameters, which define tolerable constraint violations.

Hard constraints are those that the quadratic programming solution must strictly satisfy. However, simultaneously applying hard constraints to both the control bounds and the control increments can result in infeasible solutions. To mitigate this issue, the incremental limits are treated as soft constraints by incorporating ECR. The ECR parameter is assigned to each constraint and represents the relative magnitude of an acceptable violation. This approach guarantees flexibility in the solution while maintaining system stability. Table IX presents the values of the ECR parameters for each constraint.

The scale factor property normalizes input and output variables, especially when their magnitudes differ significantly. By scaling these variables, the MPC operates with dimensionless signals, enhancing its performance. The potential benefits of scaling include the following.

- 1) *Simplified tuning and refinement*: Proper scale factors make controller tuning more intuitive and effective.
- 2) *Improved numerical conditioning*: Scaling reduces the impact of round-off errors, ensuring more accurate calculations.

TABLE X
SPECIFICATIONS OF SCALE FACTOR PARAMETERS

Signal	Type	Unit	Nominal val.	Scale factor
Plant inputs				
Q_{hc}	MV	W	0	1000
τ_{AW}	MV	%	1	1
CI1%	MV	%	1	100
CI2%	MV	%	1	100
T_o	MD	°C	25	10
I_o	MD	W/m	300	600
v_o	MD	m/s	3	6
Plant outputs				
T_a	MO	°C	20	1
T_s	UO	°C	10	1
T_r	UO	°C	20	1
T_g	UO	°C	20	0.2
T_{pv}	UO	°C	20	0.001

TABLE XI
INPUT WEIGHTS

Signal	Type	Weight	Rate weight
Q_{hc}	MV	0	0.1
τ_{AW}	MV	0	0.1
CI1%	MV	0	0.1
CI2%	MV	0	0.1

TABLE XII
OUTPUT WEIGHTS

Signal	Type	Weight
T_a	MO	1000
T_s	UO	0.1
T_r	UO	0.1
T_g	UO	0.1
T_{pv}	UO	0.1

Scale factors should ideally correspond to the span of the variable. Table X lists the scale factors for the various variables used in the system. However, the scale factors for Q_{hc} , T_a , T_s , T_r , T_g , and T_{pv} differ significantly from their spans. This adjustment ensures that the quadratic programming problem retains a positive-definite Hessian matrix, which is essential for stability and optimality.

Tuning the cost function weights is a critical step in designing a model predictive controller. These weights are assigned to both the output variables (measured and unmeasured) and the MVs. The tuning process was conducted following the guidelines provided by MathWorks [44]. The weight values for the input and output signals are presented in Tables XI and XII, respectively. For MVs, no specific target is defined, so their weight is set to 0. To prevent large increments in the MVs, their rate weight has been set to 0.1. For the output variables, priority is given to the reference tracking of T_a over the other UO. By assigning a weight of 1000 to T_a , the tracking error is significantly minimized, ensuring high precision in controlling this variable.

A look-ahead (previewing) function is added to the controller for improved performance. This enables the controller to act on MVs before changes occur in the SP or MDs. Figs. 8 and 9 illustrate the behavior with and without the look-ahead function.

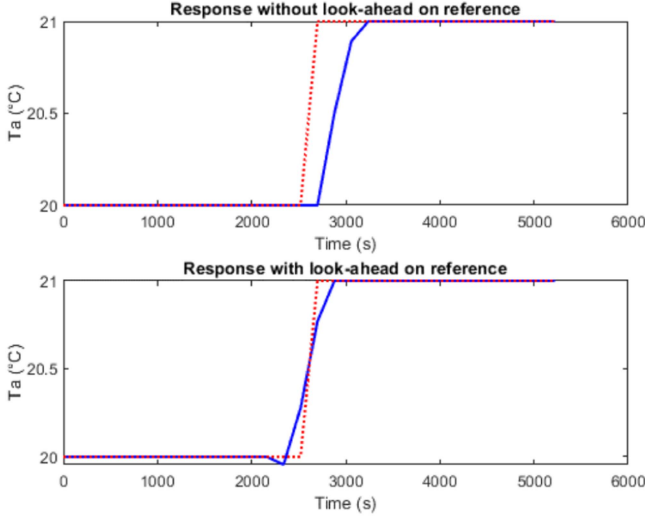


Fig. 8. Response behavior with and without look-ahead on the reference.

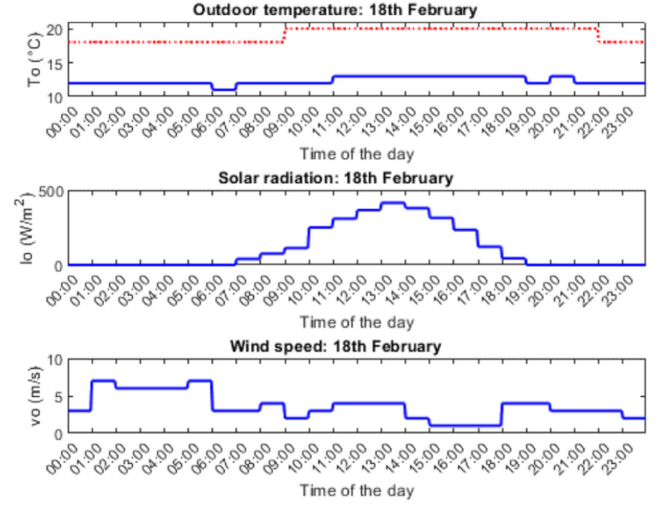
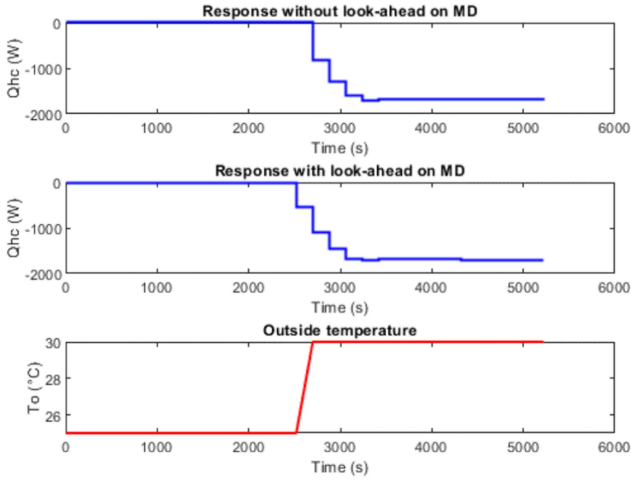
Fig. 10. Winter scenario: Q_{hc} (Winter scenario: MDs).

Fig. 9. Response behavior with and without look-ahead on MDs.

C. MPC Simulation

The MPC was tested in four separate simulation scenarios to assess its performance in varying weather conditions. All scenarios are based on historical meteorological data from Savona. The main aim of these simulations is to validate the controller's efficiency and reliability under realistic and diverse environmental conditions. The four scenarios are as follows.

- 1) *Winter*: 18 February 2021.
- 2) *Spring*: 18 May 2021.
- 3) *Summer*: 18 August 2021.
- 4) *Autumn*: 18 November 2021.

1) *Winter Scenario*: The historical data of the MDs for the winter scenario are shown in Fig. 10. Key observations include the following.

- 1) The outdoor temperature remains relatively stable throughout the day, with an average value of approximately 12 °C.
- 2) The peak shortwave solar radiation occurs at 13:00 h, with a maximum value below 500 W/m².

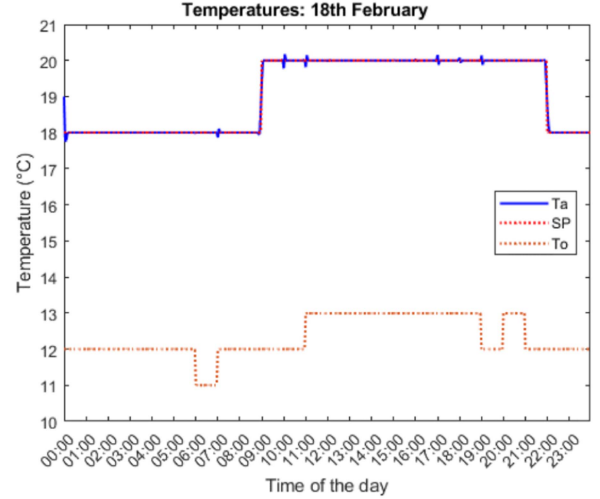


Fig. 11. Winter scenario: Ambient temperature response.

- 3) The wind speed fluctuates around 5 m/s.

The *ambient temperature* response in this scenario is shown in Fig. 11, the MVs as Q_{hc} simulation is shown in Fig. 12 and the simulation of τ_{AW} , $Cl_{1\%}$, and $Cl_{2\%}$ is shown in Fig. 13.

2) *Spring Scenario*: The historical data for the spring scenario are shown in Fig. 14. During this scenario:

- 1) the outdoor temperature rises throughout the day, reaching 20 °C at 18:00 h, with a maximum variation of 5 °C;
- 2) the shortwave solar radiation is unexpectedly low, peaking at only 200 W/m²;
- 3) wind speed fluctuates around 3 m/s.

D. Summer Scenario

The historical data for the summer scenario are shown in Fig. 15. During this scenario:

- 1) the outdoor temperature exceeds 30 °C at certain times and does not drop below 25 °C;
- 2) solar irradiance reaches a peak value of 870 W/m².

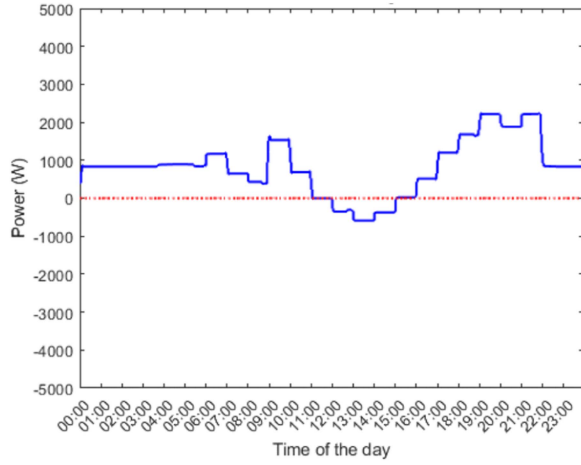
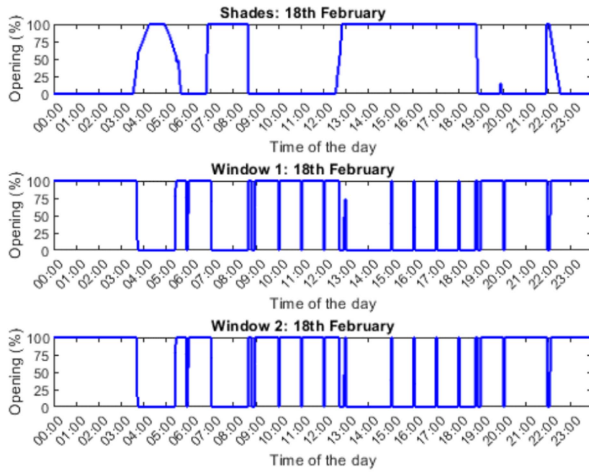
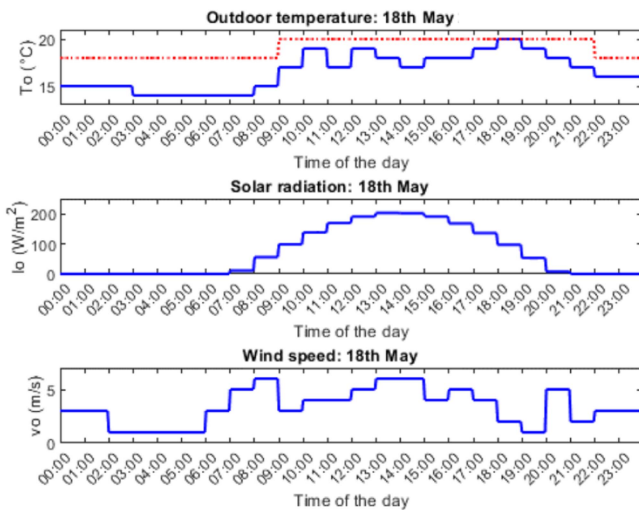
Fig. 12. Winter scenario: Q_{hc} (Heater: 18th February).Fig. 13. Winter scenario: τ_{AW} , $Cl_1\%$, and $Cl_2\%$ (Heater: 18th February).

Fig. 14. Spring scenario: MDs.

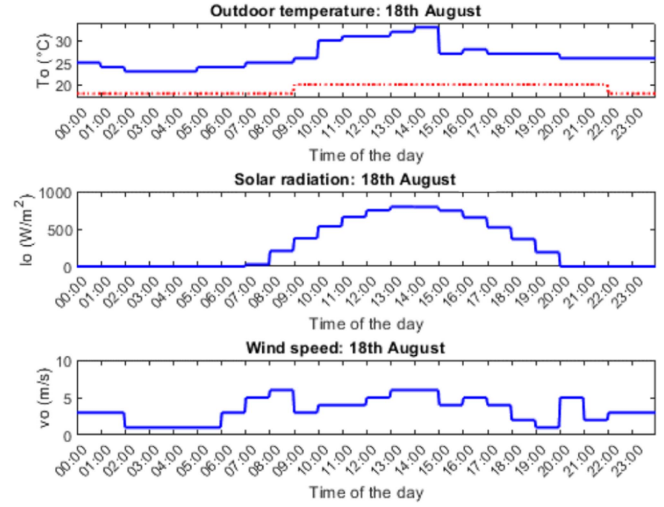


Fig. 15. Summer scenario: MDs.

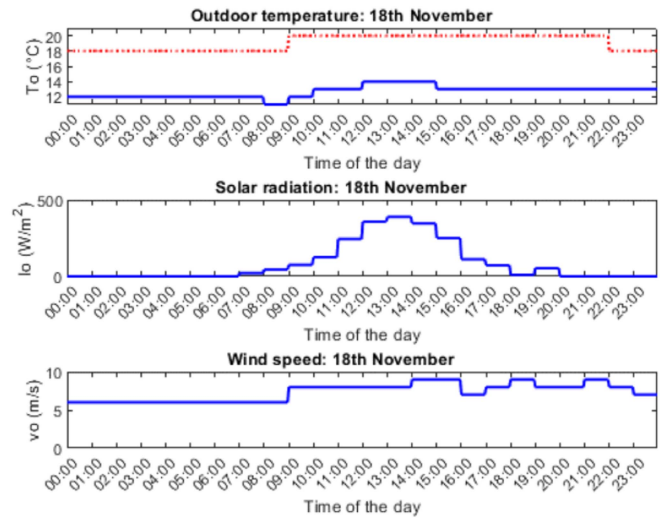


Fig. 16. Autumn scenario: MDs.

E. Autumn Scenario

The historical data for the autumn scenario is shown in Fig. 16. During this scenario:

- 1) the outdoor temperature remains stable at approximately 12 °C throughout the day;
- 2) shortwave solar radiation peaks at 13:00h with a value below 500 W/m²;
- 3) wind speed increases progressively throughout the day.

V. PHYSICAL DOMAIN

The physical domain encompasses the real-world components of the greenhouse, including sensors, actuators, and communication hardware. It interfaces directly with the DT system, as shown in Fig. 17.

The system adopts a linear bus topology where sensors and actuators communicate with a microcontroller via Modbus RTU. The microcontroller connects to a 4G gateway using TCP/IP,

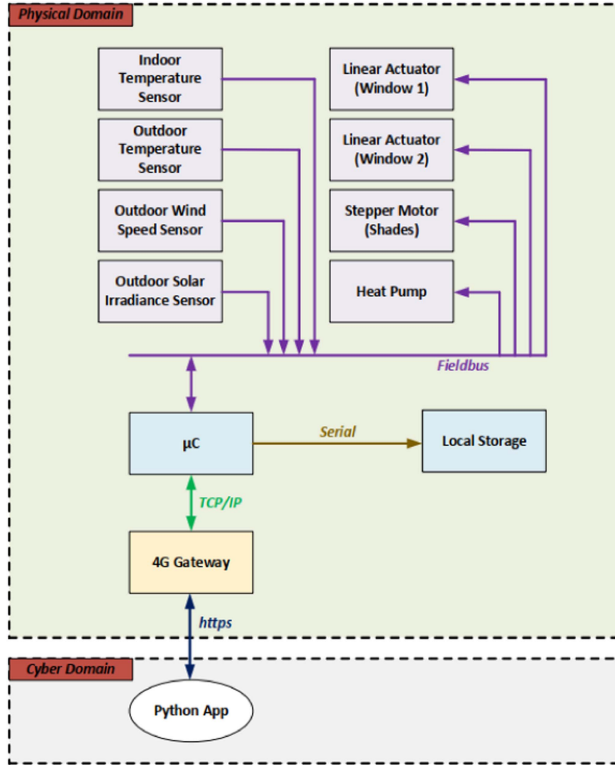


Fig. 17. Diagram of the physical domain architecture.

TABLE XIII
SENSOR REQUIREMENTS

Sensor	Use	Min range
Temperature	Indoor	[10, 35] °C
Temperature	Outdoor	[−10, 50] °C
Anemometer	Outdoor	[0, 15] m/s
PV pyranometer	Outdoor	[0, 2000] W/m ²

TABLE XIV
ACTUATOR REQUIREMENTS

Actuator	Use	Min Range	Min Precision
Linear actuator (x2)	(C11%, C12%)	[0, 0.5] m	0.05 m
Servomotor	(τ_{AW})	N/A	0.045 m
Heat pump	(Q_{hc})	[−5000, 5000] W	200 W

enabling internet access and secure data transfer. To ensure reliability during connectivity issues, sensor, and actuator data are stored in a flash memory backup. The microcontroller performs the following operations.

- 1) Reads sensor data via Modbus RTU.
- 2) Sends sensor data to the cyber domain through HTTPS.
- 3) Retrieves actuator SPs from the cyber domain via HTTPS.
- 4) Transmits SPs to actuators via Modbus RTU.
- 5) Stores sensor and actuator data locally in flash memory.

A. Hardware Requirements

The hardware requirements for sensors and actuators are listed in Tables XIII and XIV, ensuring compatibility with the DT and MPC design.

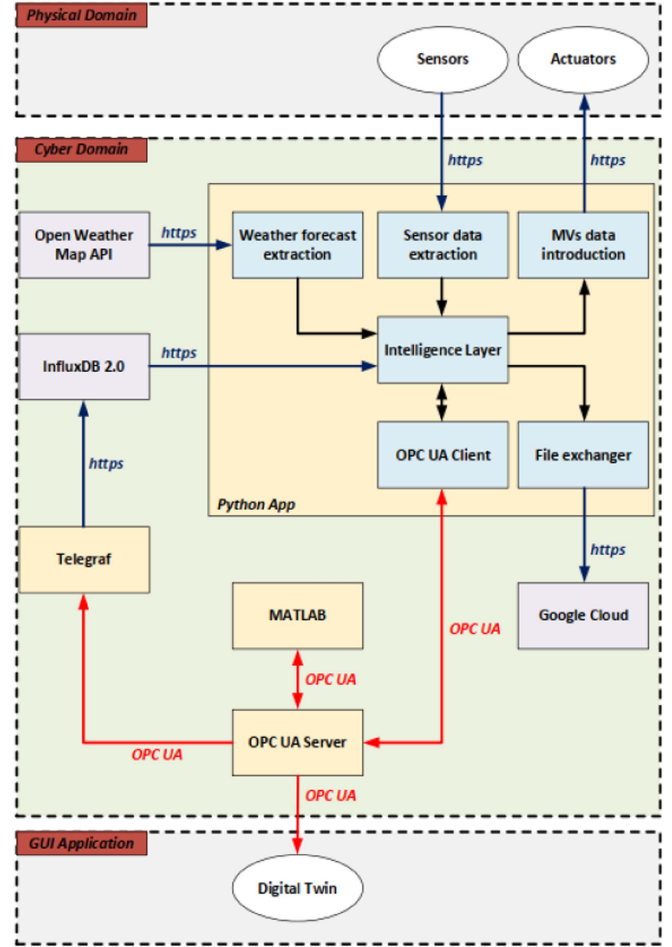


Fig. 18. Cyber domain architecture illustrating communication protocols, stand-alone applications, and cloud services.

VI. CYBER DOMAIN

The cyber domain serves as the central environment for data processing, encompassing networks, devices, software, processes, and applications that facilitate the storage, transmission, and management of information [45].

A. Cyber Domain Architecture

The architecture of the cyber domain and its data flow are depicted in Fig. 18. Based on frameworks proposed by [46], [28], and [47], the system comprises three main components:

- 1) *Communication protocols*: Define rules for data exchange.
 - a) *OPC UA*: Enables communication between stand-alone applications.
 - b) *HTTPS*: Facilitates secure data transmission between applications and cloud services.
- 2) *Stand-alone applications*: Independent programs running on a physical server.
 - a) *Python application*: Processes sensor data and retrieves weather forecasts.
 - b) *OPC UA Server*: Manages data exchange within the system.
 - c) *MATLAB*: Performs MPC calculations.

- d) *Telegraf*: Converts OPC UA data into HTTPS-compatible formats for database integration.
- 3) *Cloud services*: Provide on-demand storage and retrieval.
 - a) *InfluxDB 2.0*: A time-series database for historical data storage.
 - b) *Google cloud platform (GCP)*: Stores backup files, including CSV, XML, and MATLAB data.
 - c) *OpenWeatherMap API*: Supplies weather forecasts for greenhouse control.

B. Data Flow and Operation

The cyber domain integrates its components to ensure efficient data processing and control.

- 1) *Initialization and data collection*:
 - a) The OPC UA server establishes communication with Python, MATLAB, Telegraf, and the DT application.
 - b) Telegraf converts OPC UA data into HTTPS-compatible formats for storage in InfluxDB 2.0.
 - c) Python retrieves sensor data from the physical domain and weather forecasts from the OpenWeatherMap API. Both data sources are processed and sent to the OPC UA server.
- 2) *Controller operations*:
 - a) MATLAB reads processed sensor data, weather forecasts, and SPs from the OPC UA server.
 - b) The MPC controller calculates MV values based on the input data.
 - c) MATLAB saves the controller's state locally and sends output data to Python via OPC UA.
- 3) *Actuator control*:
 - a) Python processes MATLAB output and sends MV values to actuators in the physical domain.
- 4) *Backup and historical data*:
 - a) Python retrieves historical data from InfluxDB and sends it to GCP for backup.
 - b) MATLAB files are uploaded to GCP for long-term storage.
- 5) *DT integration*:
 - a) The DT application receives all relevant data via OPC UA for real-time monitoring and visualization.

VII. GRAPHICAL USER INTERFACE

The GUI was developed using CustomTkinter, a Python library based on Tkinter that provides modern and customizable widgets. The GUI enables real-time interaction with the DT system and is divided into two main frames.

- 1) *Navigation frame*: Located on the left side of the window, this frame allows seamless navigation between panels.
- 2) *Panel Frame*: Displays dynamic content for the current status, weather forecast, and historical data panels.

A. Core Panels and Functions

Current status panel: Displays real-time sensor and actuator data, including ambient temperature, solar irradiance, and wind speed. Graphical representations of state variables, such as soil

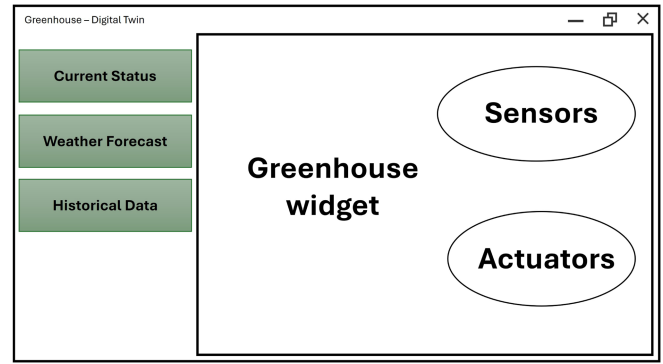


Fig. 19. GUI layout showing the Location of the frames and widgets in the current status panel.

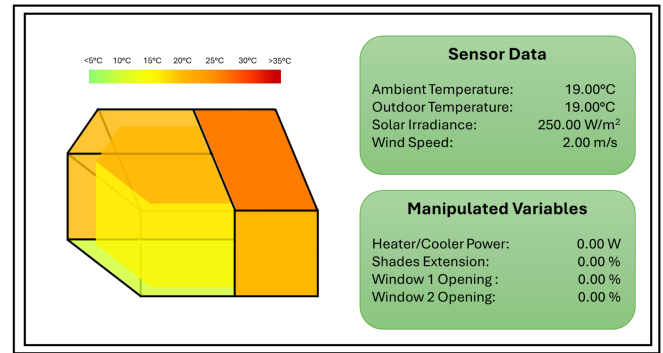


Fig. 20. Current status panel.

and PV module temperatures, are provided via a greenhouse widget that overlays temperature-specific color codes. Data are updated dynamically every 5 s.

Weather forecast panel: Retrieves weather data from OpenWeatherMap API. The panel offers:

- 1) *daily forecast*: Provides an 8-day weather outlook.
- 2) *three-hour forecast*: Displays weather predictions for the next 21 hours.

Weather maps, including air temperature (TA2), wind speed/direction, and cloudiness (CL), are dynamically updated.

Historical data panel: Allows querying and visualization of historical data stored in InfluxDB. Users can filter variables and specify time ranges using switches and combo boxes. Plots are dynamically generated and updated. Fig. 19 illustrates the layout of frames and widgets in the current status panel. It highlights the user interface's organization of components, guaranteeing intuitive navigation and efficient data visualization. Fig. 20 presents the current status panel, highlighting its functionality for displaying real-time data such as ambient temperature, solar irradiance, and wind speed.

VIII. RESULTS

Based on historical weather data from Savona, the DT system for greenhouse management was tested in four seasonal scenarios (winter, spring, summer, and autumn). The simulations demonstrated the robustness of the MPC in maintaining optimal climatic conditions. For example, during the summer scenario,

when outdoor temperatures exceeded 30 °C and solar irradiance peaked at 870 W/m², the controller effectively adjusted shading (τ_{AW}) and ventilation ($C_{11}\%$, $C_{12}\%$) to mitigate heat build-up. Similarly, in winter, with relatively stable outdoor temperatures around 12 °C and low solar irradiance, the heating system (Q_{hc}) was used effectively to maintain the desired indoor temperature. These findings highlight the system's ability to respond dynamically to changing climatic conditions, guaranteeing consistent greenhouse performance while meeting operational constraints.

Although this study focuses on the case of Savona, the system architecture is designed to be adaptable to other types of greenhouses and various climatic conditions. Indeed, the modular nature of the sensors and actuators means they can be reconfigured to suit different structural configurations or environmental requirements. In addition, recalibration of MPC models for specific regional weather conditions could enable generalization to wider applications. Further work will aim to validate this adaptability by carrying out tests in different geographical and operational contexts.

In addition, the integration of real-time sensor data and weather forecasts into the MPC via the cyber domain enabled accurate forecasts and timely adjustments. Furthermore, the use of flexible constraints, implemented via ECR, has ensured feasible solutions even under difficult conditions, such as rapid temperature fluctuations during transition seasons.

In this study, discrepancies between the DT model and actual conditions, such as sensor interference or unmodeled dynamics, were taken into account. Although these deviations were found to be generally negligible due to frequent model updates, their effects could become significant in scenarios involving faulty sensors or extreme weather conditions. The system mitigates these risks by using regular calibration cycles and exploiting real-time feedback from the physical domain. Other improvements, such as adaptive modeling techniques, could help to remedy residual discrepancies.

Moreover, by reproducing system behavior and presenting data for visualization and analysis, the DT provided real-time feedback. This enabled any deviations or anomalies to be quickly detected and corrected. The integration of predictive weather data via APIs also enhanced the controller's anticipatory capabilities, resulting in smoother operation and fewer corrective measures.

The proposed methodology has demonstrated its flexibility in adapting to various greenhouse architectures. For instance, window placement or shading mechanisms may differ according to design, requiring optimization of sensor placement and actuator configurations. These adaptations are simple within the system's modular framework, allowing scalability and customization for various structural configurations.

The system's performance underlines its ability to seamlessly integrate various components. Communication between the cyber and physical domains, facilitated by OPC UA protocols, ran smoothly, ensuring efficient data transmission. The intelligence layer enabled historical data analysis and backup management, improving system resilience and reliability.

Indeed, the environmental impact analysis highlighted server electricity consumption as the dominant factor contributing to

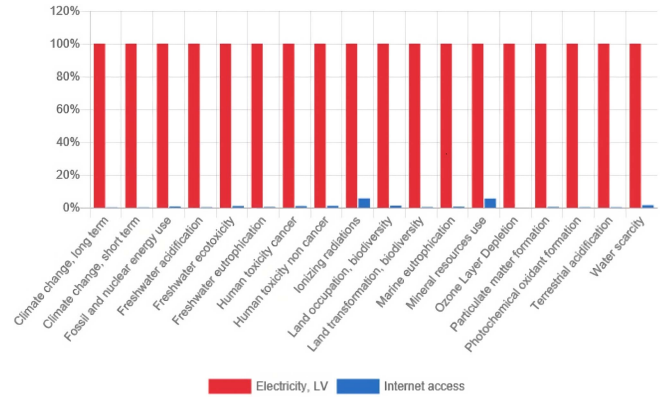


Fig. 21. LCA relative results.

TABLE XV
DETAILED PROJECT COST BREAKDOWN

Category	Item	Cost
Engineering Labour	Gross salary	9375.00 €
	Common contingency contribution	2212.50 €
	Training contribution	56.25 €
	Unemployment contribution	628.13 €
	FOGASA contribution	18.75 €
	Occupational accident contribution	140.62 €
	Subtotal (Engineering Labor)	12,434.25 €
Software Licenses	MATLAB	840.00 €
	Symbolic Math Toolbox	420.00 €
	Model Predictive Control Toolbox	840.00 €
	Control System Toolbox	480.00 €
	Industrial Communication Toolbox	520.00 €
	Subtotal (Software Licenses)	3,100.00 €
Cloud Services	OpenWeatherMap Developer Plan	800.00 €
	Storage GCP	0.12 €
	Operative Charges GCP	0.25 €
	Subtotal (Cloud Services)	800.37 €
Overall Total Cost		16,334.62 €

the system's ecological footprint. According to life cycle assessment (LCA) data, electricity consumption accounted for over 99% of CO₂ equivalent emissions, dwarfing the contributions of Internet use. Fig. 21 illustrates the relative environmental impacts of server energy consumption compared to internet access. These results underline the need for energy-efficient hardware solutions, such as low-power servers or the integration of renewable energies, to improve system sustainability.

In economic terms, the project's total cost was estimated at € 16 334.62, with the largest expenses coming from engineering labor (76%), and software licenses (19%). While MATLAB's robustness and extensive capabilities justified its use, the associated license costs had a significant impact on the budget, as shown in Table XV. The transition to open-source tools, such as Python and its libraries, could significantly reduce software expenditure; however, it comes at the cost of increased development time and expertise. Similarly, optimizing the use of the OpenWeatherMap API to balance costs and functionality is an avenue worth exploring.

IX. CONCLUSION

This work highlights the performance of a DT integrated with an MPC for smart and efficient agricultural operations.

Simulation results validate the effectiveness of the proposed framework in achieving its objectives in various operational scenarios.

The MPC simulations demonstrate accurate SP tracking while maintaining stability in controlling environmental variables. The system successfully adapted to scenarios with fluctuating weather conditions by adjusting the MVs, guaranteeing optimal growing conditions for the crops. In addition, seasonal analysis highlighted the controller's robustness in both summer and winter scenarios, ensuring energy efficiency and consistent environmental control.

The results underline the potential of integrating DT technology with advanced control systems, such as MPC, to revolutionize intelligent agriculture. The proposed framework lays the foundations for scalable and sustainable smart farming solutions by improving operational efficiency, reducing costs, and minimizing environmental impact.

Limitations of DTs in greenhouse systems include expensive starting costs, complicated integration, and reliance on precise data, which might result in inaccurate forecasts. The accuracy and responsiveness of simulations can be diminished by biological variability, external environmental influences, and computational constraints. To optimize the advantages of DTs in greenhouse operations, careful planning and administration are also required due to scalability constraints, ethical considerations surrounding data ownership, and the possibility of system breakdowns.

Although this study is mainly based on historical data, the methodology is well-suited to real-time deployment in operational greenhouses. During initial evaluations, a comparison was made between the results produced by the MPC and deterministic models based on real measurements. The findings showed that the results were approximately equivalent, validating the accuracy of the MPC-based approach. This consistency supports the use of MPC data as a reliable proxy for real-world results in the current study. Future research will focus on integrating real measurements in a live greenhouse environment to further validate and refine the DT model dynamically under real operating conditions. This step will enable a more comprehensive assessment of the system's performance and adaptability.

Considering environmental and economic aspects highlights the need for strategic decisions aimed at balancing sustainability and profitability. The use of energy-efficient servers and open-source software offers opportunities to reduce both ecological and financial burdens. However, maintaining system functionality and performance remains crucial to successfully implementing DT technologies in intelligent agriculture.

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